Effect Sovereignty

System Calls as the Constitutional Interface for Reality-Touching Effects

The computational core remains effect-pure; interaction with external reality becomes governed, inspectable, and replayable.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Modern systems increasingly blur the boundary between computation and the world. Graphical interfaces, network calls, storage persistence, distributed RPC, sensor inputs, and time itself are often mediated by layered runtimes and frameworks that treat world interaction as library functionality rather than constitutional crossing. This diffusion obscures where authority resides and how external effects are licensed.

This paper argues that any operation whose effects escape the computational core must be modeled as a constitutional boundary crossing. We introduce the concept of **effect sovereignty**: the principle that world-touching operations must traverse explicit authority contracts with structured outcomes including execution, simulation, deferral, and refusal. The computational core remains effect-pure; interaction with external reality becomes governed, inspectable, and replayable.

Building on the constitutional syscall model, we generalize the boundary abstraction beyond classical kernel traps. We provide a taxonomy of reality-touching effects, formalize a minimal effect-boundary interface, and demonstrate how this architecture enables temporal portability, deterministic replay, and multi-authority adjudication across heterogeneous execution environments.

The result is not syscall expansion but architectural clarification: world interaction becomes a constitutional act, and authority evaluation becomes part of the execution substrate.

---

1. The Core–World Divide

At the heart of any computing system lies a distinction that is often implicit but rarely formalized:

- The computational core: deterministic transformation of internal state.
- The world: any domain whose effects persist beyond that core.

The computational core can be reasoned about using program semantics, type systems, and formal verification. The world resists such closure. It includes:

- Persistent storage
- External devices
- Network communication
- Time and scheduling
- Human interaction through GUI
- Distributed systems beyond the current process boundary

Traditional architectures entangle these domains. Libraries initiate network calls. GUI frameworks allocate windows. Time is queried as a convenience function. In doing so, world interaction becomes indistinguishable from internal computation.

This conflation obscures a structural fact: any world-touching operation is a crossing of jurisdiction.

---

2. Defining Reality-Touching Effects

We define a **reality-touching effect** as any operation whose outcome:

1. Persists beyond the computational core's internal state
2. Influences or depends upon external authority domains
3. Cannot be fully simulated without reference to an external contract

Examples include:

- Writing to disk
- Sending network packets
- Emitting sound or drawing pixels
- Reading sensor input
- Querying wall-clock time
- Committing distributed transactions

These operations are not merely functions. They alter or consult domains not owned by the local computation. As such, they must be governed.

---

3. From Library Call to Constitutional Crossing

In classical systems, world-touching effects are often embedded in libraries that internally issue system calls. Authority is assumed to be resolved elsewhere, typically in the kernel.

This indirection creates opacity. The boundary exists, but it is obscured by layering.

We propose instead:

> Any reality-touching effect must be expressed as a constitutional boundary crossing, evaluated by an explicit admissibility function.

Under this model, world interaction is not delegated to opaque frameworks. It is declared, adjudicated, and logged at a governed interface.

The boundary may be:

- A kernel
- A runtime
- A sandbox
- A replay engine
- A distributed authority coordinator

What matters is not its identity but its explicit contract.

---

4. Effect Sovereignty

Effect sovereignty is the principle that the computational core does not self-authorize external impact.

The core produces proposals for world interaction. The boundary adjudicates them.

Formally, we define:

$$\text{EffectBoundary} : \text{Effect} \times \text{Authority} \to \text{Outcome} \times \text{Reason}$$

Where:

- Effect encodes the intended world-touching operation
- Authority encapsulates capabilities and contracts
- $\text{Outcome} \in \{\text{execute}, \text{simulate}, \text{defer}, \text{refuse}\}$
- Reason is structured metadata explaining the decision

This model ensures:

- Execution is explicit
- Simulation is legitimate
- Deferral is tracked
- Refusal is first-class and auditable

The computational core remains effect-pure; all impurity is mediated.

---

5. Taxonomy of Constitutional Effects

Reality-touching effects can be categorized by domain:

5.1 Persistence Effects

- File writes
- Database commits
- Log append operations

These effects alter long-lived state.

5.2 Communication Effects

- Network send/receive
- Inter-process communication
- Distributed RPC

These effects cross process or machine boundaries.

5.3 Temporal Effects

- Reading system time
- Scheduling tasks
- Sleeping or waiting

Time introduces non-determinism and must be governed.

5.4 Interface Effects

- GUI rendering
- Input event handling
- Audio/video output

These effects mediate human interaction.

5.5 Sensor and Actuator Effects

- Reading hardware sensors
- Triggering actuators
- Issuing control signals to physical systems

Each class demands explicit jurisdiction evaluation.

---

6. Multi-Authority Adjudication

In practice, world interaction may traverse multiple authorities:

- Replay engine
- Runtime sandbox
- Operating system
- Hardware interface

Adjudication composes sequentially:

$$\text{Admissibility}_{\text{chain}} = A_1 \circ A_2 \circ \cdots \circ A_n$$

Each authority may:

- Approve
- Modify
- Simulate
- Defer
- Refuse

Failure at any layer results in structured refusal.

This composition ensures that authority evaluation is layered yet explicit. No layer silently self-authorizes beyond its declared scope.

---

7. Deterministic Replay and Audit

When all reality-touching effects cross constitutional boundaries, each decision can be recorded:

ConstitutionalLogEntry = (Effect, Outcome, Reason, ContractID, Timestamp, AuthorityID)

This log enables:

- Deterministic replay through simulation of prior decisions
- Auditability of refusal patterns
- Post-hoc analysis of authority behavior

Replay engines can serve as primary adjudicators, simulating effects under recorded authority contracts. The system thus decouples meaning from immediate execution.

---

8. Kernel as One Authority Among Many

Under this architecture, the kernel is not eliminated but repositioned.

It becomes:

- A possible authority in the adjudication chain
- Not the ontological anchor of world interaction

A WebAssembly runtime, a replay engine, or a distributed governance layer may act as primary adjudicator. The kernel's role becomes one participant in a broader constitutional landscape.

This shift mirrors the evolution from monolithic kernels to microkernels and capability systems, but extends further by making effect evaluation itself first-class.

---

9. Costs and Tradeoffs

Structured adjudication introduces overhead:

- Contract management
- Structured reason propagation
- Logging and schema maintenance

These costs are analogous to those observed in capability-based systems and formally verified kernels. They represent deliberate tradeoffs in favor of explicit authority and reproducibility.

Effect sovereignty does not eliminate complexity. It renders it inspectable.

---

10. Architectural Implications

Treating reality-touching operations as constitutional boundaries yields several consequences:

- The computational core can be formally reasoned about independently of external effects
- Undefined behavior at the world interface is replaced by structured refusal
- Simulation becomes a first-class operational mode rather than a hack
- Execution environments can evolve without redefining effect semantics
- Authority drift becomes detectable rather than implicit

This model complements the broader Substrate Migration pattern: authority evaluation itself descends into substrate enforcement, replacing advisory discipline with governed execution.

---

11. Conclusion

World interaction is not a side effect of computation. It is a constitutional act.

By treating reality-touching effects as governed boundary crossings, we preserve the purity of the computational core while making authority explicit at every point of external impact. Execution becomes one possible adjudicated outcome among several; refusal becomes legitimate; simulation becomes principled.

In such systems, world interaction is no longer an implicit privilege granted by position. It is a licensed crossing of jurisdiction.

The machine does not merely execute instructions. It mediates its contact with reality under contract.

Corpus-Wide Reference Positioning

This paper argues that modern systems increasingly blur the boundary between computation and the world. Its nearest external comparison fields are security, governance, observation. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Unforgeable capabilities can constrain exercise of authority while still leaving allocation, delegation, revocation, and social legitimacy incorrectly specified. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Telemetry can be complete at the instrumented boundary while omitting the causal variable, real-world effect, or authority question that matters. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant

work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Jerome H. Saltzer and Michael D. Schroeder. "The Protection of Information in Computer Systems." *Proceedings of the IEEE* 63(9), 1975, pp. 1278-1308. DOI: 10.1109/PROC.1975.9939. https://doi.org/10.1109/PROC.1975.9939 **Relationship:** Foundational protection principles used as a comparison point for least authority, mediation, and inspectability.

[**E2**] Mark S. Miller. *Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control*. PhD dissertation, Johns Hopkins University, 2006. http://www.erights.org/talks/thesis/ **Relationship:** Predecessor object-capability treatment of authority and compositional security.

[**E3**] Gerwin Klein et al. "seL4: Formal Verification of an OS Kernel." *SOSP '09*, 2009, pp. 207-220. DOI: 10.1145/1629575.1629596. https://doi.org/10.1145/1629575.1629596 **Relationship:** Predecessor for machine-checked implementation correspondence and capability-based kernel design.

[**E4**] Robert N. M. Watson, Simon W. Moore, Peter Sewell, and Peter G. Neumann. *An Introduction to CHERI*. University of Cambridge Technical Report UCAM-CL-TR-941, 2019. DOI: 10.48456/tr-941. cl.cam.ac.uk source **Relationship:** Adjacent capability architecture demonstrating unforgeable hardware authority and compartmentalization.

[**E5**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E6**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E7**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

Ableman corpus

[**A1**] Adam Ableman. *Receipt-Bound Capabilities: Membrane Tokens for Governed Interaction with Reality.* Ableman Corpus current edition; work identity `PAPER.PUB.18601631`;
source SHA-256
`6DE46F821594AB11BFB7F8373368C452BD89E5BD4EB172B88D9467E87CF74C33`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *The Integrity Enforcement Layer.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-505307E9E6571B63`;
source SHA-256
`F858AB32E0937B796353CAE573DD4183FDF4F179A3E1E7E40C4E0C6BF1AB0018`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Constitutional Syscalls.* Ableman Corpus current edition; work identity `PAPER.PUB.18737229`;
source SHA-256
`5F14356180812262778F8574C292F4F6B560993042E5358B02EAB9CF0F7DE14C`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-1F9BFB2BF8370226`;
source SHA-256
`AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Ableman Constitutional Governance.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Constitutional Theory.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D2D97C4042FD281F`;
source SHA-256
`FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governance Engineering.* Ableman Corpus current edition; work identity `PAPER.PUB.18749870`;
source SHA-256
`372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.18737323